

Test Plan – Order Generator Feature

Overview

The Order Generator is responsible for automatically creating randomized orders based on predefined difficulty levels and system rules. Since the feature contains both deterministic and random elements, testing focuses on validating boundary conditions and system behavior under load.

This initial test plan includes:

- Two boundary tests (edge-case validation)
- One stress test (system stability under repeated execution)

These tests follow Unity Test Runner guidelines, where boundary tests validate input limits and stress tests simulate high-load execution.

Boundary Test 1: Ingredient Count Limits

Purpose

To verify that the number of ingredients in generated orders remains within the defined limits for each difficulty level.

Description

Each difficulty level has strict ingredient boundaries:

- Easy: 1–2 ingredients
- Medium: 3–4 ingredients
- Hard: 5–6 ingredients

This test evaluates values at, below, and above these limits.

Test Cases

- Easy difficulty with 0 ingredients (below minimum)
- Easy difficulty with 1 ingredient (minimum boundary)
- Easy difficulty with 2 ingredients (maximum boundary)
- Easy difficulty with 3 ingredients (above maximum)
- Hard difficulty with 4 ingredients (minimum boundary)
- Hard difficulty with 6 ingredients (maximum boundary)
- Hard difficulty with 7 ingredients (above maximum)

Expected Results

- Orders must only contain ingredient counts within valid ranges
- Values outside the defined limits must not occur
- Any invalid values must be corrected or rejected by the system

Boundary Test 2: Empty or Null Order Handling

Purpose

To verify that the system correctly handles cases where order generation fails or produces invalid output.

Description

If the order generator produces an empty or null order, the system must trigger error handling and replace it with a predefined default order.

Test Cases

- Input: Empty order
- Input: Null order
- Input: Valid generated order

Expected Results

- The system must not crash or throw unhandled exceptions
- Empty or null orders must trigger error handling
- A default order must be generated when a failure occurs
- The system must continue execution normally after recovery
- Default orders must only appear in failure scenarios

Stress Test: Continuous Order Generation

Purpose

To evaluate system performance and stability under extended execution and repeated order generation.

Description

The system will continuously generate, process, and validate orders over a large number of cycles to simulate real runtime conditions.

Test Scenario

- Execute 500 consecutive order generation cycles

- Each cycle includes:
Generate Order → Process Order → Validate Order

Expected Results

- No runtime exceptions or crashes occur
- All generated orders remain structurally valid
- The system continues operating without interruption
- Orders are generated sequentially without skipping or duplication errors
- Default orders appear only when generation failures are triggered
- System performance remains stable (no freezes or abnormal slowdowns)

Conclusion

This initial test plan ensures that the Order Generator behaves correctly at boundary conditions and remains stable under heavy load. Together, these tests validate both the correctness and robustness of the feature before full integration testing.